A Blockchain Enabled P2P Bidding Communication System Integrated with AI and Cloud



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DEPARTMENT OF ELECTRICAL ENGINEERING

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Abstract

The growing trend of EV adoption faces a backlash due to limited charging infrastructure and high energy costs, especially in third world countries where charging infrastructure is in infancy. This project proposes a peer-to-peer (P2P) energy trading platform that uses artificial intelligence (AI), block chain technology and cloud integration. The platform supports role-switching that allows users to switch roles whether as buyers or sellers, real time location fetching, path navigation and engaging in decentralized energy bidding. A secure block chain is implemented as a decentralized database that operates as a digital ledger securely registering peer-to-peer (P2P) transactions across a distributed network without deploying central authority. The platforms aims to support at least ten prosumers with future work including GUI development. Benefits include clean energy adoption.

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Chapter 1

Introduction

1.1 Introduction

In recent years, we have witnessed that in other countries there are a sufficient amount of charging stations, and EVs have increased rapidly. But when we look at developing countries, their adaption of EVs is very low. This is because of lower number of charging stations when compared with other countries such as China and India. We have very a very limited number of charging stations– for instance, all over Pakistan we have 8-13 charging stations. This is a major setback for the adoption of EVs in developing counties (Pakistan). The low number of charging stations causes range anxiety and inconvenience. This is the major reason why people don't adopt EVs. For instance, there is a vehicle that is currently charged with more energy than is required to travel. After reaching the destination, it still has some remaining energy which can be utilized to reach a third point. However, this often isn't possible due to limited availability of charging stations in third world countries. As a result, the remaining energy goes unused, which results in energy mismanagement. Currently, most systems in this time are decentralized, which means they don't use any middlemen, making the transactions more secure and trustworthy. However, in centralized authority, there is a whole organization that deals with your transactions. This project aims to address challenges by cre ating a trading platform that will deal with prosumers. This trading platform will introduce and increase the productivity of EVs in developing countries by letting the EV owners to buy and sell energy to each other. This will also remove inconve nience of limited charging stations, and all the energy transactions will be secured through a block chain with a decentralized database. The use of block chain in our

project will help build trust between prosumers. This project encourages green technology adoption

1.2 Main Parts

1. P2P Communication
2. V2V Communication
3. V2G Communication

1.2.1 P2P Communication

Peer-to-peer communication is a type of communication where a user can act as both as a client and a server without relying on a central authority, each participant or peer has equal capabilities and responsibilities. P2P communication offers enhanced security and privacy. We can apply the concept of P2P communication to energy sharing where energy consumers can also act as energy providers, which plays a crucial role in enabling the direct, decentralized exchange of energy without requiring a third person. This type of energy sharing can make power sharing more efficient by enabling consumers (vehicle, grids) to become active participants in the energy market. Advancing the idea of P2P communication, V2G and V2V communication make it possible for EVs to share energy either with grid or directly with other vehicles

1.2.2 V2V Communication

In the context of energy sharing between electric vehicles (EVs) using V2V (Vehicle to-Vehicle) communication, P2P (Peer-to-Peer) communication can enable vehi cles to directly exchange energy without relying on an infrastructure. When an EV with a full battery detects that another EV has a low battery, P2P commu nication allows them to negotiate and directly transfer energy. This is done using

vehicle-to-vehicle communication protocols that allow both vehicles to securely exchange data, including their battery levels, location, and energy needs

1.2.3 V2G Communication

In a vehicle-to-grid configuration, EVs are able to both return energy to the grid when required and receive energy from the grid to charge their batteries. This bidirectional energy transfer is made possible by P2P communication between EVs and the grid. This lowers electricity prices and helps balance energy supply and demand, particularly during peak hours. Additionally, it converts users into active energy contributors and can supply backup power during blackouts

1.3 Blockchain

Block Chain is a decentralized technology that enables secure, transparent and tamper proof data transactions. In a decentralized system, there is no central authority or a single point of control; instead data is distributed across a network of computers known as nodes, where each participant has equal power to access and verify information. At its core, block chain functions as a distributed ledger, meaning it is a digital record for transactions that is shared among all participants in the network unlike traditional ledgers which are centralized (controlled by one entity like banks). A distributed ledger exists across multiple computers, ensuring transparency and security and preventing any single entity from manipulating the data. In block chain, data is organized in blocks each containing a list of trans actions. These blocks are linked together in chronological order to form a chain, which is why it is called block chain. Each contains a cryptographic hash of the previous block, making it impossible to alter past transactions without changing all subsequent blocks. The concept of public and private block chain refers to how the block chain network is structured in terms of accessibility, control and governance. Both types are decentralized but they differ significantly in their in tended use cases and the level of openness or permission required for participants.

A public block chain is an open, permission less network where anyone can join participate and validate transactions. It is completely decentralized and does not require permissions from any central authority to access or contribute to the block chain. Public blockchains offer high transparency and security but are slower and are high power compared to private ones. Bitcoin is the most common cryptocur rency built on a public blockchain, which is often considered the best type for open and trustless environments.

This chapter introduced the motivation behind the growing need for decentralized, peer-to-peer energy trading platforms in response to the rising adoption of electric vehicles (EVs) and the challenges posed by limited charging infrastructure, particularly in developing regions. It outlined the core concept of blockchain technology, highlighting the distinction between public and private blockchains and their respective trade-offs in transparency, security, and performance. The chapter also discussed Vehicle-to-Vehicle (V2V) and Vehicle-to-Grid (V2G) communication paradigms, which form the backbone of decentralized energy exchange between EVs and the power grid, along with the broader communication infrastructure required to support real-time, secure, and trustless energy transactions. Together, these concepts establish the foundation for the proposed peer-to-peer energy trading platform discussed in this thesis.

The next chapter, Chapter 2: Literature Review, builds on this foundation by examining existing research and implementations related to blockchain-based energy trading, V2V/V2G communication frameworks, and AI-driven trading mechanisms. It critically reviews prior work in the field, identifies existing gaps and limitations, and positions the contributions of this project within the broader research landscape.

Chapter 2

Literature Review

2.1 Introduction

To understand the project's fundamental principles, a literature review is done. It involves reading through earlier studies that have relevance to the idea of our project. This stage helps us get an understanding of our project, which helps with project design and execution. After reading a number of research papers, the research articles that follow have helped us improve our project idea

2.2 Previous Work

A comprehensive study on energy trading and finance using blockchain technology 2025 Miglani et al. (2025) [1] provide a comprehensive overview of how blockchain technology revolutionizes energy trading and finance by enabling secure, decentralized, and transparent peer-to-peer (P2P) transactions. This research contributes to understanding the role of decentralized energy systems in transforming energy transactions, finance, and sustainability. The authors highlight how blockchain might promote the involvement of prosumers, facilitate the integration of renewable energy sources, and make new financial products like energy banking and green bonds possible. It also highlights the gaps that need to be addressed through research and responsible policy actions.

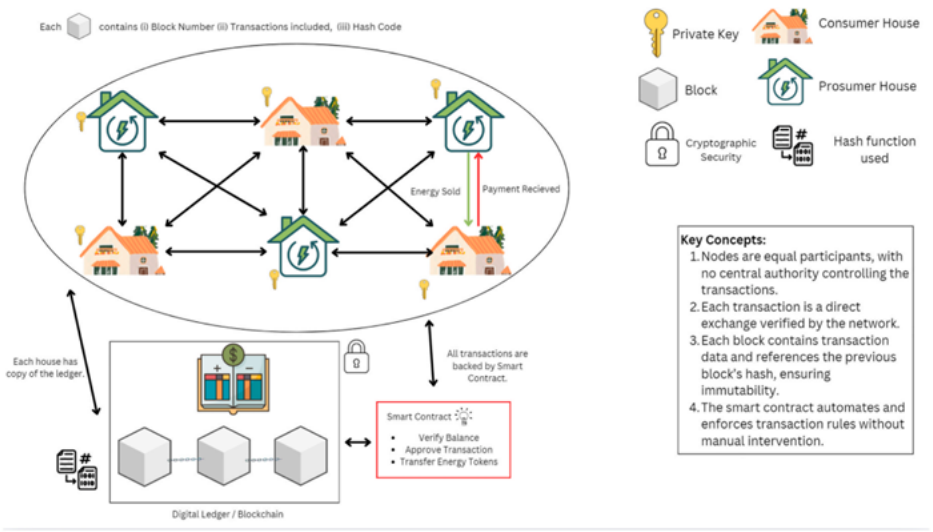


Fig. 1. A diagram explaining the use of Blockchain for Energy Trading and Finance.

Figure 2.1: A diagram explaining the use of Blockchain for Energy Trading and Finance.

Fair Energy Trading in Blockchain: Inspired Smart Grid: Technological Barriers and Future Trends in the Age of Electric Vehicles 2024 Qazi et al. (2024) [2] conduct an in-depth exploration of energy trading in smart grids (SGs) enhanced by blockchain, emphasizing its growing importance amidst the rising adoption of electric vehicles (EVs). This research explores the challenges and opportunities arising from the integration of electric vehicles into smart grids. It highlights the importance of blockchain for secure, transparent transactions, and addresses the engineering, policy, and cybersecurity issues that must be resolved for effective EV and smart grid integration. This research also mentions future trends and technological advancements that will shape the evolution of smart grids in the coming years.

Comparison between Blockchain P2P Energy Trading and Conventional Incentive Mechanisms for Distributed Energy Resources—A Rural Microgrid Use Case Study Aoun et al. (2024) [3] compare blockchain-based peer-to-peer (P2P) energy trading with traditional billing mechanisms like Net Metering (NEM) and

Feed-in-Tariff (FiT) within a rural microgrid setting. This research evaluates and compares the economic impacts of three energy trading mechanisms (P2P, NEM, and FiT) within a rural microgrid. This study showed that P2P energy trading offers greater flexibility and higher economic benefits than traditional systems. Additionally, it provided valuable insights for policymakers and stakeholders interested in optimizing rural energy systems through innovative energy trading mechanisms, potentially improving the efficiency and financial sustainability of energy systems.

The research presented in Scientific Reports (2024) [4] explore blockchain consensus mechanisms specifically designed for peer-to-peer electricity trading systems, addressing the growing need for decentralized energy markets in the context of renewable energy integration. This study introduces a comprehensive secure user verification system that authenticates participants before allowing them to engage in energy transactions, ensuring that only legitimate users can access the trading platform. The framework implements automatic smart contracts that execute trades without manual intervention, significantly reducing transaction time and eliminating human error while ensuring that all agreed-upon terms are fulfilled precisely as programmed.

This September 2024 publication in Scientific Reports [5] examine the broader applications of blockchain technology in peer-to-peer energy markets, with particular emphasis on clean energy initiatives and the role of distributed ledger technology in facilitating sustainable energy transitions. The research demonstrates how blockchain enables direct trading between energy producers and consumers, eliminating the need for intermediaries such as utility companies or energy brokers, thereby reducing transaction costs and empowering individual participants to control their energy resources. The system ensures transparent transactions through immutable ledger records, allowing all participants to verify trades independently without relying on a central authority, which builds confidence in the fairness and accuracy of the trading process.

Published in Scientific Reports in August 2022, [6] present a lightweight blockchain framework designed to address computational efficiency in peer-to-peer energy trading, recognizing that traditional blockchain implementations often suffer from excessive resource consumption that makes them impractical for resource-constrained environments. The framework achieves fast transaction processing through optimized consensus algorithms that reduce the computational burden compared to conventional proof-of-work mechanisms, enabling near-instantaneous energy trades that are essential for responsive energy markets where supply and demand fluctuate rapidly. Unlike traditional platforms that charge transaction fees to cover operational costs and incentivize miners, this system eliminates extra costs through an innovative fee structure, making it economically attractive for small-scale energy producers such as residential solar panel owners.

This chapter reviewed existing research on blockchain-based peer-to-peer energy trading, V2V/V2G communication frameworks, and AI-driven trading mechanisms, highlighting key contributions as well as gaps in scalability, cost-efficiency, and decentralized trust. Several studies demonstrated the feasibility of decentralized energy markets but struggled to balance transaction speed with security guarantees, while others achieved strong performance at the cost of high computational overhead unsuitable for resource-constrained environments. Across the reviewed literature, recurring challenges included high transaction fees that discourage small-scale prosumers, limited support for real-time bidding, and a lack of integrated communication frameworks tailored specifically to electric vehicle energy exchange. These insights directly inform the design choices made in this project. The next chapter, Chapter 3: Methodology, presents the system architecture and implementation approach adopted to address these gaps. In particular, the gaps identified around transaction throughput and fee structures shaped the design of the block append and chain update mechanisms detailed in the following chapter, while the limitations in real-time communication motivated the WebSocket-based notification system implemented later in the platform.

Chapter 3

Methodology

3.1 Introduction

Design and implementation of an energy trading platform that fully supports flexible role switching between buyer and seller modes, enables real-time location fetching for accurate tracking, incorporates dynamic path navigation for optimized energy delivery routes, and facilitates a transparent and competitive energy bidding mechanism. Design and implementation of a blockchain system that is specifically tailored to securely record and manage peer-to-peer (P2P) energy transactions, ensuring immutability, decentralization, and transparency in the ex change of energy units between participating entities. Design and implementa tion of a robust communication system that allows seamless interaction between electric vehicles, the network infrastructure, prosumers, and other EVs, enabling efficient data exchange, transaction coordination, and real-time communication for effective energy trading.

3.2 Block Diagram

This block diagram shows how our exchanging framework works between electric vehicles.

1. Each EV user (buyer/seller) updates key info: battery level, bidding rate, and location
2. This data is sent to the GUI (Graphical User Interface) for interaction.
3. The GUI sends all updated values to the Platform Update module
4. From here, the data flows to the Cloud, which acts as the central controller.
5. The cloud shares transaction info with Miners, who create a New Block.
6. Latest transaction details are validated and stored in the Blockchain.

7. The blockchain is connected to a Database for record-keeping and future ref erence.
8. Finally, all verified updates are pushed back to the platform to reflect real-time data

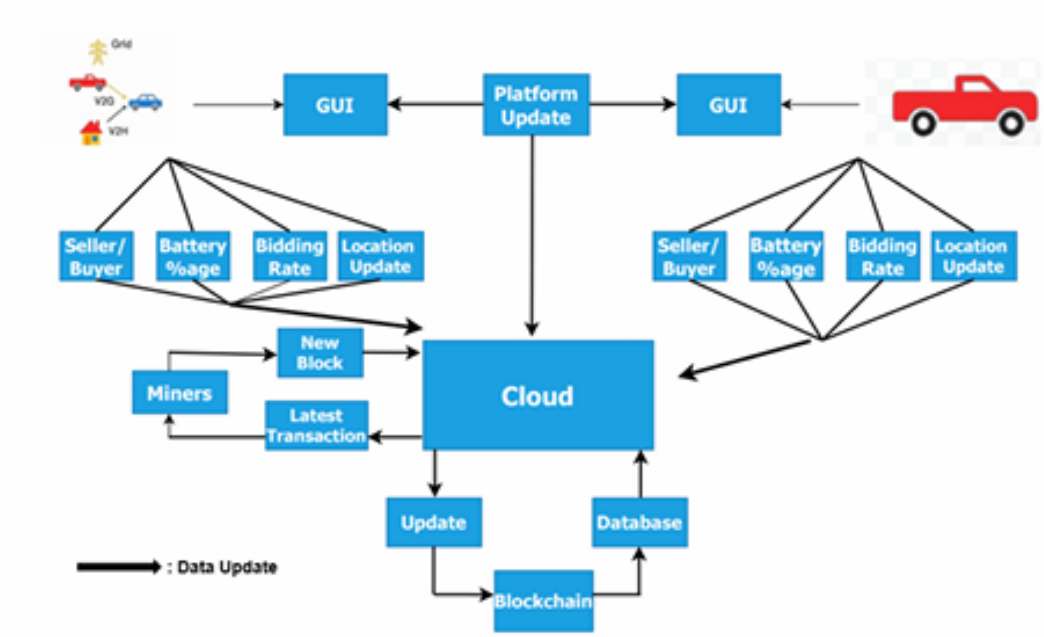


Figure 3.1: Block Diagram

As illustrated in Figure 3.1, the system architecture connects sellers and buyers through a centralized cloud layer that coordinates platform updates, bidding rates, and location data in real time. Miners validate new transactions and append them to the blockchain, ensuring that updates pushed to both buyer and seller GUIs remain synchronized. This design enables secure, low-latency communication between all participants while maintaining a tamper-proof record of every transaction.

3.3 Flow Chart

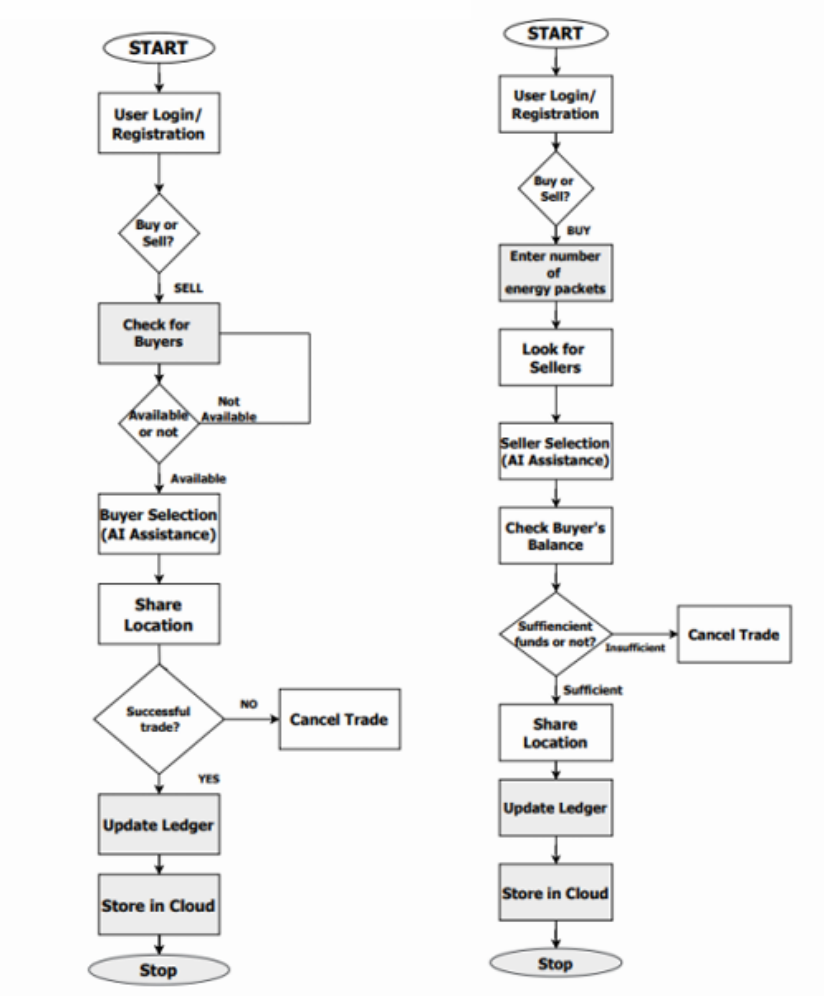


Figure 3.2: Flow Chart

3.3.1 Buyer

1. The process starts when a user logs in and selects a role (buyer)
2. The buyer enters the number of energy packets they want to purchase.
3. The system searches for available sellers
4. Once sellers are found, the buyer selects one again with the help of AI for optimal match
5. The system then checks the buyer's current account balance.

6. If the buyer has insufficient, the system cancels the trade.
7. If sufficient, the system shares the seller's location for physical interaction
8. In then updates in the digital ledger
9. Finally, the transaction is stored in the cloud, and the process ends

3.3.2 Seller

1. The process starts when a user logs in and selects a role (seller)
2. The system checks for buyers who are currently available.
3. If buyers are available, the seller proceeds to select one with the help of AI to choose the nearest and low-priced seller.
4. The system then checks if the trade is successful or not
5. If unsuccessful, the system cancels the trade
6. If the trade is successful, the system proceeds to share location details for physical interaction
7. The system then updates the transaction in the digital ledger
8. Lastly, the transaction is stored in the cloud, and the process ends

3.4 Project Deliverables

Our project will deliver three main things. First, a working energy trading platform that can support at least 10 users who act as both buyers and sellers (prosumers). Second, basic communication hardware that lets these users connect and trade. This also includes microcontrollers and communication modules. And finally, we'll build a blockchain system that safely records all trades and keeps everything updated in real time.

Together, these deliverables form a complete end-to-end pipeline, from the physical hardware that enables user interaction to the underlying blockchain layer that ensures every transaction is securely recorded. The following sections detail the design and implementation of each of these components.

This chapter presented the methodology adopted for the development of the proposed peer-to-peer energy trading platform, including the system block diagram, flow chart, and the step-by-step trading process from bid placement to transaction completion. It also outlined the core project deliverables and discussed how the project aligns with relevant UN Sustainable Development Goals (SDGs) by promoting affordable clean energy, innovation, and sustainable communities. With the overall approach and deliverables established, the next chapter.

Chapter 4

Design Procedure and Details

4.1 User Information

The user information module manages everything from initial registration to final payment processing. Each person who joins our platform gets their own unique identity and secure access credentials. We've designed it so that buyers and sellers can safely exchange energy while keeping their financial transactions protected. Here's how the whole system operates:

4.1.1 User Registration

When someone wants to join E-Drive, they start by creating an account through our mobile interface. They pick a username and password, just like any other app. But here's where we go further, we also ask them to connect their bank account while signup. Why? Because when they actually buy or sell energy later, the money needs somewhere to go. Once registration is complete, our system generates a unique ID for that person. We store all this information in two places: MongoDB Atlas (our cloud database) and the blockchain. This dual storage approach gives us speed and security, MongoDB handles the day-to-day operations while

4.1.2 User Authentication (Sign-In Process)

Once you've registered, getting back into the app is straightforward. You simply enter your username and password like you would on any platform. Behind the scenes, credentials against what's stored in our MongoDB database will be verified. If everything matches, you're in. The app then takes you to your personal dashboard where you make a key decision if user wants to sell energy today or buy it? This flexibility is important because your needs might change

4.1.3 Unique User ID and Role Assignment

Each individual utilizing our platform is assigned a distinct identifier, analogous to a digital fingerprint. This identification assists us in monitoring the purchasing and selling activities, while also ensuring that all transactions are systematically organized. Upon your decision to engage in the sale of energy, it will be necessary to update certain aspects of your profile: the available battery percentage, the price you are requesting per kilowatt-hour, and your current geographical location (which is determined via GPS technology). Conversely, if you are in the position of a buyer, you have the opportunity to explore local sellers, evaluate their pricing, ascertain the quantity of energy they are offering, and select the option that aligns best with your preferences.

4.1.4 User Profile

Your profile serves as a repository for personal information pertinent to transaction activities. We request fundamental contact information, specifically your name, telephone number, and email address. More crucially, you will provide your banking information in this section: the financial institution you utilize, your account number, and the name of the account holder. At this juncture, one might question the necessity for a buyer to access a seller's banking information. Indeed, when a transaction is executed, the buyer must ascertain the destination for the payment. Consequently, upon your acceptance of an offer, your banking details will be disseminated to the other party in order to facilitate the monetary transfer.

4.1.5 Transaction Record and Security

One great thing about our system is that every single energy transaction is forever etched in the blockchain; it cannot be deleted, it cannot be faked, and it cannot be tampered with post facto. The transaction history is transparent: the number of kWh you bought or sold, the price you paid or received per unit, and so on.

We have even implemented multi-factor authentication for sensitive actions such as changing your password or your bank details, as this will prevent unauthorized access to your account.

4.2 Vehicular Information

4.2.1 Seller/Buyer Role Selection

After you register and log in, we just ask, do you want to buy or sell? If you are a seller, your vehicle is posted to the list of vehicles that buyers can view; if you are a buyer, you can view the list of current sellers. But this data is not locked on your phone. It is pushed to our central server on Render.com that serves as the central brain of the operation, monitoring every vehicle in the network and its availability.

4.2.2 Battery Percentage Monitoring

Your EV already has a Battery Management System that tracks your remaining charge, and we just connect to that and send the data to our server where it may be updated every few seconds or every couple of minutes (frequently enough to be accurate but not so much that it drains your phone battery). There is also a bit of smart logic to this: if you are selling energy but your battery is dangerously low, our system will take you off the seller list and suggest that you switch to buyer mode (it seems logical that you should not be selling energy if you do not have extra to spare).

4.2.3 Rate of Sell

Selling is free market, and you can charge whatever you want per kWh based on demand, location, or how badly you need to sell, and then upload your rate to our server, where everyone can see it and the buyer can shop around and not be locked into one person and see what every single seller is charging. We also have

smart contracts that are triggered when a deal is made that lock in the agreed price and will not allow for exceptions.

4.2.4 Location Update

We rely on location tracking for our system, and we use GPS modules such as the NEO-6M to accomplish this. The position of your vehicle is updated continuously and coordinates are sent to our server. Why is location so important? Because energy transmission loses energy, the further you are from your trading partner, the more energy is lost in the transmission. Our algorithm, therefore, actively attempts to match people who are close to each other.

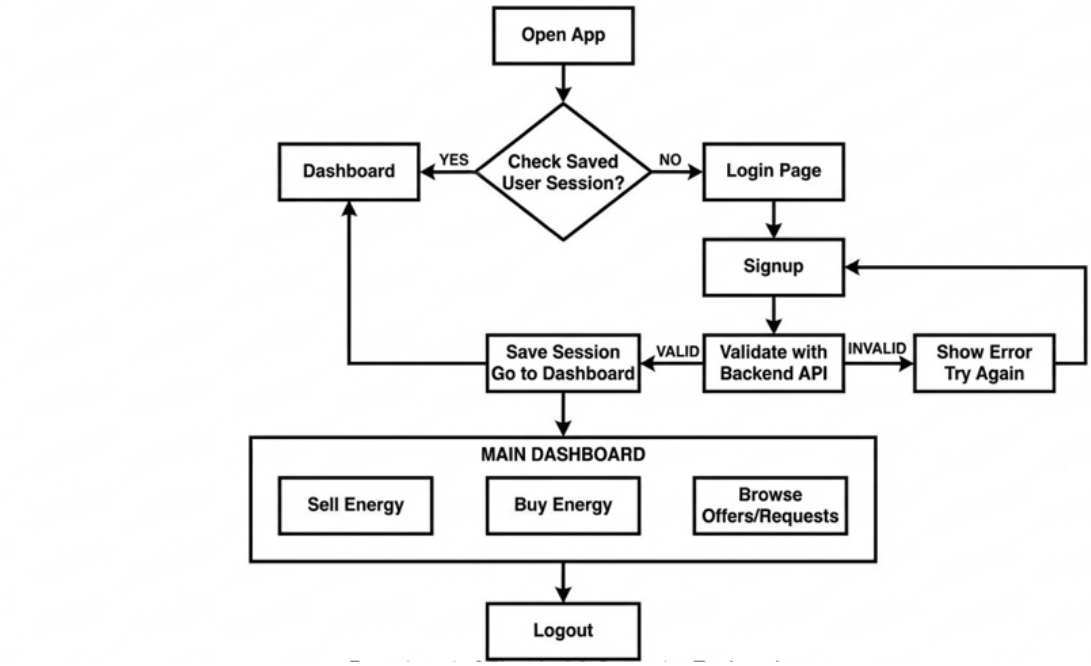


Figure 4.1: Authentication and Main Navigation

4.2.5 Blockchain Integration

You may have heard of blockchain, but for our app, it is the security model: each energy transaction is put in a ”block” of data, which is cryptographically sealed and connected to the previous block, so that once it is added to the chain, it cannot

be changed without invalidating all the blocks that came after, thereby creating an unchangeable record of what was sold to whom, when, and for how much. We're using a decentralized approach, which means there's no single authority controlling everything instead, the system itself enforces the rules. This is perfect for peer-to-peer trading between EV owners because nobody has to "trust" a central company; you can verify transactions yourself. Each block contains five key components:

1. **Block Number (N-1, N, N+1):** The block number indicates a block's precise location in the blockchain ledger. It ensures that the transaction sequence can be tracked chronologically by giving each block a unique identification. This arrangement is essential for preserving the ledger's structure, verifying the data's historical flow.

2. **Timestamp:** Each block includes the date and time it was created and added to the blockchain, which provides an immutable record of when transactions occurred, as well as a means for determining the sequence of events if there is ever a disagreement or network issue.

3. **Last Block Hash:** The smart part of blockchain is that each block has a hash (or digital fingerprint) of the previous block, creating a chain in which each block is mathematically linked to the previous one. If someone tries to go back in time and alter some data, the hash of that block will change, and the next block with the old hash value will immediately reveal that the data has been altered, and the chain will be broken. This linking is what makes blockchain essentially tamper-proof.

4. **Transactions:** The actual transaction data lives here; all deals within a certain time window go into the block transaction section. What we store: the basics: the seller ID, the buyer ID, the amount of energy exchanged, the price agreed upon, and digital signatures from each party to confirm they both authorized the trade. If these transactions make it into a block and

that block becomes part of the blockchain, they are immutable; they cannot be deleted, altered, or disputed.

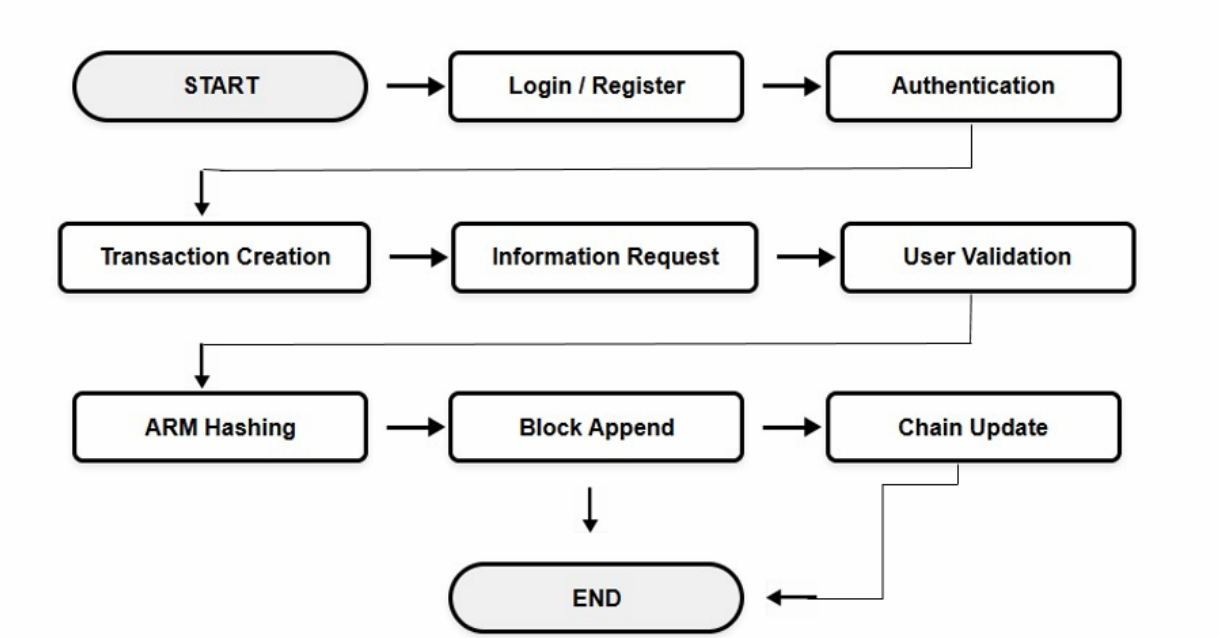


Figure 4.2: Blockchain Flow

This chapter presented the detailed design and implementation procedure of the proposed peer-to-peer energy trading platform, covering the system architecture, the E-Drive blockchain process flow, and the underlying mechanisms for transaction creation, validation, and chain updates. It explained how ARM hashing, block appending, and chain updates work together to ensure that once a transaction is recorded, it becomes immutable and tamper-proof. With the system design and implementation now established, the next chapter, Chapter 5: Results and Discussion, presents the outcomes of testing the platform and evaluates its performance against the project's objectives.

Chapter 5

Results and Discussion

They tested from multiple angles: the mobile app functionality, backend API responses, database synchronization, the matching algorithm's decision-making, and the overall user experience. Each component had to function on its own and seamlessly interface with the other components of the system. In this chapter, you will see the payoff of that work: screenshots of the actual app, MongoDB collections, API response logs, and explanations of how each piece fits into the overall energy trading ecosystem.

5.1 Mobile Application Implementation

React Native with Expo was a good choice for our app because we can write code once and deploy it to both iOS and Android, which saves us a tremendous amount of time versus building two different apps. The frontend is responsible for what users see and interact with: registration, login, buyer/seller toggling, view of available energy offers, and initiating the transaction. The app is structured in a fairly standard navigation pattern: Users are presented with a login/registration screen (new users can create accounts here, while existing users can sign in), then after authentication they are presented with the main dashboard, where the buyer/seller decision takes place (we implemented it as a simple toggle or selection screen, where picking "Seller" yields one interface, and picking "Buyer" yields something else).

For sellers, the interface shows fields to input their selling rate (price per kWh) and the amount of energy they want to offer. There is also real-time display of their current battery percentage and location. When they submit an offer, it immediately gets sent to the Flask backend, which stores it in MongoDB and makes it visible to potential buyers.

The screen on the buyer side displays a list of all sellers in the area, with their GPS coordinates, asking price, and the amount of energy they are offering; the buyer can scroll through the options, compare prices, and choose the one who offers the best deal, at which point the app determines distance using the Haversine formula and initiates the transaction using the backend API.

Getting the navigation flow right took several iterations. Initially, we had too many screens and users were getting lost. We simplified it down to the essential paths:

Login → Dashboard → Buyer/Seller Selection → Transaction Screen → Confirmation

This streamlined flow reduced the number of taps required to complete a trade and significantly improved usability during testing, as users were able to complete the buying or selling process without confusion or unnecessary back-and-forth between screens.

5.2 Flask Backend and API Integration

All of the logic occurs in the backend, which we developed with Flask, a lightweight but powerful Python web framework, and exposes several API endpoints that the mobile app calls to execute various operations. Here are the key endpoints we implemented:

1. **User Registration (/api/register):** Takes username, password, and bank account details from the signup form. Creates a new user document in MongoDB with a unique ID. Returns success/failure status back to the app.

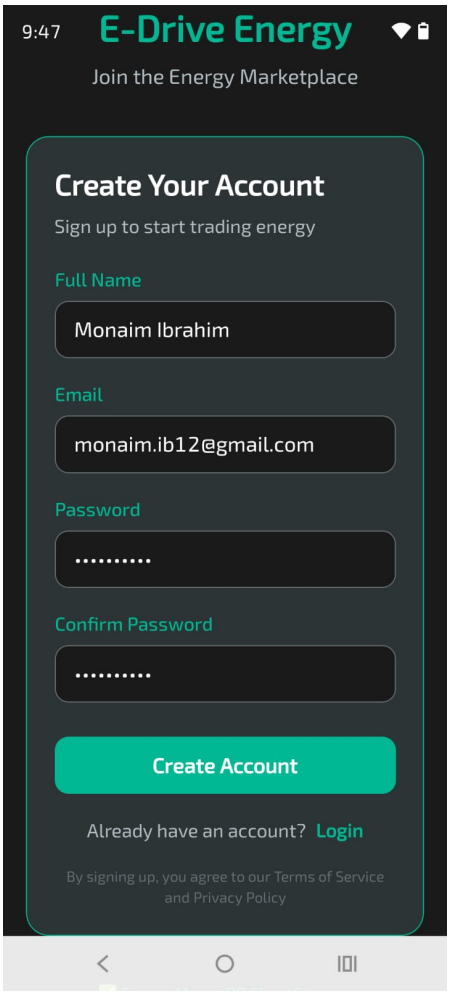


Figure 5.1: SignUp Page

2. **User Login (/api/login):** Accepts username, password, and bank account information from the signup form, creates a new user document in MongoDB with a unique ID, returns success/failure status to the app, verifies credentials against stored records, if successful, generates an authentication token and sends it back, and the app saves this token and sends it along with all subsequent API calls for verification.

   Together, the SignUp and Login endpoints form the entry point into the platform, ensuring that only authenticated users can access trading features. Once logged in, the session token returned by the backend is used to authorize all further interactions, including offer submissions and transaction

confirmations.

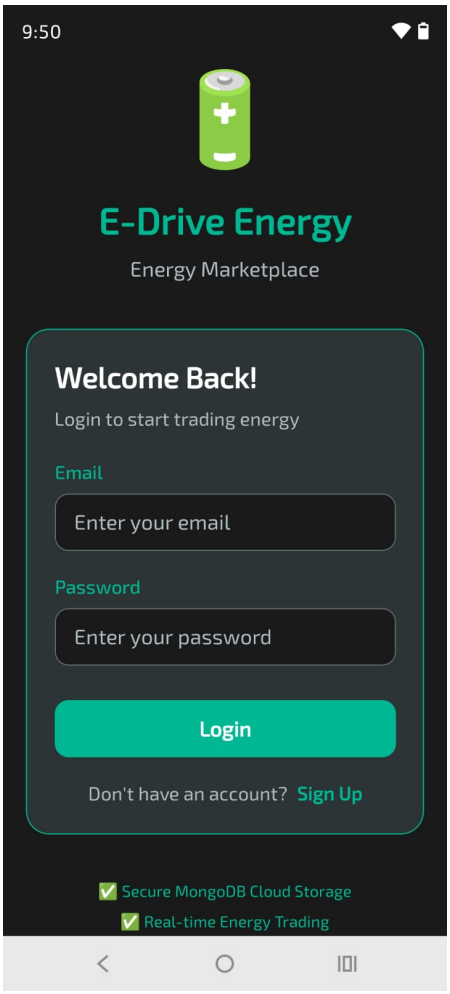


Figure 5.2: Login Page

3. **Submit Seller Offer (/api/sell/offer):** Invoked when a buyer chooses a seller. Stores a transaction with the buyer ID, seller ID, energy amount, price, and timestamp, and updates both users' statuses in the database.

   This endpoint is central to the trading process, as it captures all the details needed to track a transaction from initiation through to completion, ensuring that both the buyer and seller's records remain synchronized in real time.

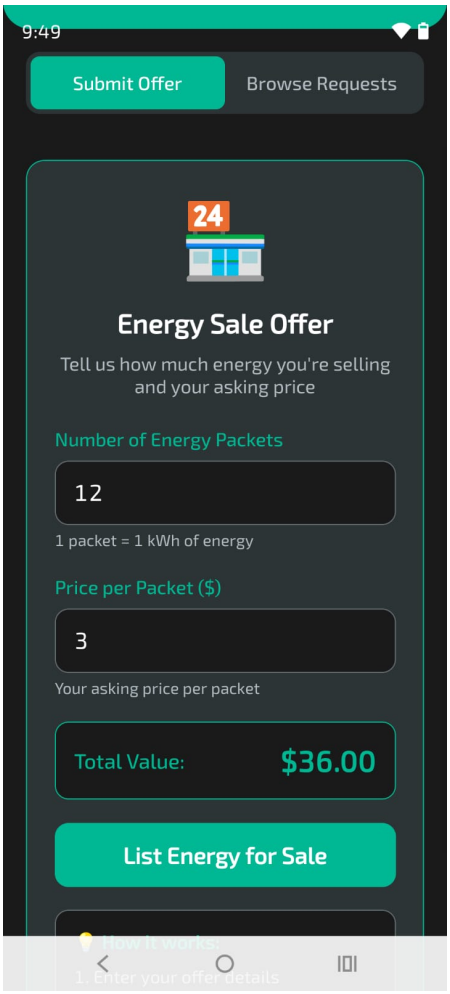


Figure 5.3: Seller Interface

4. **Get Available Sellers (/api/buy/sellers):** Returns all active seller offers with details, which the app displays in the buyer interface. We included optional query parameters to filter by distance or price range. Takes in the rate from the seller, the amount of energy, the battery percentage, and the GPS coordinates. Stores the seller offer in the database with status active so that other users can see this offer.

5. **Initiate Transaction (/api/transaction/create):** Called when a buyer selects a seller. Creates a transaction record with buyer ID, seller ID, energy amount, price, and timestamp. Updates both users' statuses in the database.

5.3 MongoDB Atlas Database Structure

MongoDB Atlas is our cloud database solution. We chose it because it's flexible with schema design, scales well, and integrates easily with Flask. The database is organized into several collections, each serving a specific purpose:

1. **Users Collection** All user account data is stored here. With fields like userId (unique identifier), username, passwordHash (plain passwords are not stored), email, phone number, bankAccountDetails (account number, bank name, account title), and created, each document represents a single user. at the timestamp. We organized it so that all user-related information is kept in one location for convenient access.


```
_id: ObjectId('693d98969ecbe2ee6b49d3a1')
user_id : "634209eb-0b36-45be-abcb-af044033ace0"
name : "Monaim Ibrahim"
email : "monaim.ib12@gmail.com"
password : "pakistan12"
wallet_balance : 1000
created_at : "2025-12-13T16:47:18.649875"
session_token : "e79be016-1fea-4204-91cd-5e1fe3a4d989"
last_login : "2025-12-13T16:47:58.620275"
role : "buyer"
```


Figure 5.4: User Database

2. **Sellers Collection** Listings for active sellers go here. SellerId (which identifies the user), location (latitude and longitude), battery percentage, energy available (in kWh), price per kWh, status (active/inactive), and timestamp are all included in each document. A new document is inserted when a seller offer is created. The status becomes inactive when the offer expires or is fulfilled.

   This structure allows the platform to quickly query all active sellers within a given area and retrieve their current offer details without scanning unrelated records, which keeps the matching and bidding process fast and efficient even as the number of users grows.

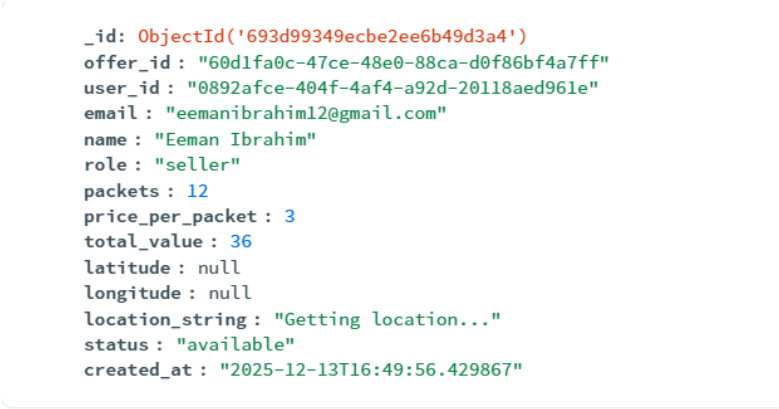


Figure 5.5: Seller Information

3. **Transactions Collection** Every completed energy trade creates a transaction record. Fields include transactionId, buyerId, sellerId, energyTransferred, pricePerKwh, totalAmount, buyerLocation, sellerLocation, distance, timestamp, and status (pending/completed/failed). This gives us a complete audit trail of all marketplace activity.



Figure 5.6: User.json

4. **Blockchain Collection** We store block-like records here for transaction verification even though this isn't a conventional blockchain implementation. BlockNumber, previousHash, merkleRoot, currentHash, transactions array, and timestamp are all present in every document. This gives security and transparency the immutability we desire.

MongoDB's flexibility really helped during development. When we realized we

needed to add a field for user ratings or modify the transaction structure, we could just update the schema without complicated database migrations. The Atlas cloud hosting means the database is accessible from anywhere, which is essential since our backend is on Render and the mobile app can be running on devices all over the world.

5.4 User Interface and Experience Validation

We tested extensively on both Android and iOS to ensure everything worked properly across platforms. While the React Native framework handles most of the heavy lifting for cross-platform compatibility, we still needed to tackle some platform-specific issues.

For Android, we tested using the emulator in Android Studio and a Samsung device. Our biggest issue was with GPS permissions, as Android 10 and above require explicit runtime permissions for location access; we implemented a permission request flow that prompts users to enable location tracking when opening the app for the first time, as location data is necessary for the matching algorithm to function.

Since we do not have a physical iPhone, we tested on the Xcode simulator for iOS. Permissions are handled a little differently in iOS: the system will prompt the user the first time the app attempts to use location, and we simply had to ensure that my Info.plist file contained the appropriate usage descriptions outlining why we require location access.

The UI went through several design iterations: early versions were functional but dull (white backgrounds with black text and basic buttons), we refined it by adding colors that align with the energy/sustainability theme (greens and blues), enhanced button styling (padding and shadows), and ensuring text was legible across screen sizes, and added loading spinners for network requests so users are not left staring at a blank screen wondering if the app froze.

Navigation flow testing revealed that users sometimes got confused about their

current role (buyer vs seller). we solved this by adding a persistent indicator at the top of the screen showing which mode they're in, plus a quick toggle button to switch roles without navigating back to the dashboard.

5.5 Functional Testing and Results

To validate that everything works end-to-end, we ran through several test scenarios simulating real user interactions:

1. **Test Scenario 1: Complete Transaction Flow** Test account setup: Seller (User A) and buyer (User B). User A posted an energy offer: 10 kWh at 15 PKR/kWh with a fake location near Centaurus Mall. User B browsed available sellers, saw User Aś offer at the top of the list (due to matching algorithm), and initiated a transaction. The backend created a transaction record, updated the statuses of both users, and stored the data in MongoDB. Time from buyer selection to database confirmation: approximately 2.3 seconds.

2. **Test Scenario 2: Multiple Sellers Comparison** Created five different seller accounts with varying prices (ranging from 12 to 20 PKR/kWh) and different locations around Islamabad/Rawalpindi. As a buyer, we could see all five options ranked by the matching score. The cheapest seller who was also relatively close appeared first, while an expensive seller far away ranked last. Verified that the Haversine distance calculations were accurate by cross-checking with Google Maps distances.

3. **Role Switching** Started as a buyer, browsed sellers, then switched to seller mode without logging out. Posted my own offer, which immediately appeared in the database. Switched back to buyer mode - my posted offer correctly didn't show up in my own buyer list (you can't buy from yourself). Verified that role transitions happened smoothly without data loss or navigation errors.

4. **Test Scenario 4: Network Interruption Handling** Intentionally disabled WiFi while the app was trying to load seller data. The app displayed an appropriate error message instead of crashing. Re-enabled WiFi and hit the retry button - data loaded successfully. This confirmed that my error handling works and the app fails gracefully rather than breaking completely when network issues occur.

5.6 System Performance Analysis

1. **API Response Times:** Most requests are handled by the Render-hosted Flask backend in 200–500 milliseconds. Creating transactions takes about 350 milliseconds, retrieving seller lists takes about 400 milliseconds (including database query time), and login requests take about 300 milliseconds on average. Users don't notice delays under 500 ms, so these are reasonable speeds for a mobile application.

2. **Database Query Performance:** MongoDB Atlas effectively manages queries. It takes between 50 and 80 milliseconds to retrieve all active sellers (usually 5 to 10 records during testing). Because the username field is indexed, user authentication queries are even faster, taking only 30 to 40 milliseconds. Although performance is good for now, we may need to add more indexes or use pagination as the database expands.

3. **Mobile App Load Time:** React Native apps typically take two to three seconds to launch initially. Because we're only switching components rather than reloading the entire application, subsequent screen transitions are almost instantaneous (less than 100 ms). To speed up initial render times, we implemented lazy loading for images.

Table ?? summarizes the key specifications of the E-Drive platform across size and scale, user handling, cloud infrastructure, and security.

Category	**Specification**	**Value**
Size & Scale	Target concurrent users	10+ prosumers (initial deployment)
	Transaction throughput	12–20 TPS under concurrent load
	Block verification time	Under 100 ms for 100-block chains
	WebSocket notification latency	Under 35 ms
User Handling	Authentication	Email/password with SHA-256 hashing
	Role switching	Dynamic buyer/seller toggle per user
	Session management	Token-based session authentication
	Real-time updates	Socket.IO room-based per-user broadcasting
Cloud Infrastructure	Backend hosting	Render.com (Flask, Gunicorn)
	Database	MongoDB Atlas (cloud-hosted NoSQL)
	Blockchain service	Custom hybrid blockchain microservice
	API communication	REST endpoints over HTTPS
	Frontend platform	React Native (Expo), cross-platform iOS/Android
Security	Hashing algorithm	Custom ARM-256 (blockchain proof)
	Encryption	AES-256-GCM with PBKDF2 key derivation
	Wallet system	ECDSA-based wallet addresses

Table 5.1: E-Drive System Specifications

5.7 Technical Challenges Encountered and Resolved

Challenge 1: MongoDB Connection Issues: Initially, Flask backend kept losing connection to MongoDB Atlas, especially after periods of inactivity. The error message said something about connection pooling. Solution: we configured the MongoDB client to use connection pooling properly and increased the connection timeout values. Also added retry logic so if a database operation fails, it automatically retries up to 3 times before giving up.

Challenge 2: Navigation State Management: Challenge 2: Navigation State Management: Users would toggle between buyerto seller mode, but the app would sometimes display outdated data fromthe old role due to improper state management when switchingroles. Solution: Properly managed state using React hooks (useStateand useEffect), so when the user switches roles, the app clears thepertinent state variables and fetches new data from the backend.

Challenge 3: GPS Coordinate Formatting: Challenge 3: GPS Coordinate Formatting: The mobile app was sendingGPS coordinates as strings (”33.7050”, ”73.0680”) but my Haversine formula was expecting numbers, which would cause the calculation to fail silently, returning NaN. Solution: Converted types in the backend to parse coordinate strings into floats before doing any math and added validation to reject invalid coordinates.

5.8 Hybrid Blockchain Implementation

A sophisticated implementation that combines private security and public transparency is the hybrid blockchain system. This chapter provides thorough findings from the creation and testing of a dual-chain architecture that makes use of proof-of-work consensus techniques, AES-GCM encryption, and unique ARM256 cryptographic hashing. Secure user authentication, encrypted transaction processing, unchangeable record-keeping, and real-time balance management across dispersed nodes are all successfully demonstrated by the system.

User registration and authentication processes, secret key generation and management, transaction creation and validation, blockchain mining with adjustable difficulty, cryptographic hash verification, and access control mechanisms were all included in the testing. To guarantee that security, performance, and dependability meet production-grade standards, every subsystem underwent a thorough evaluation.

5.9 System Architecture and Components

The hybrid blockchain architecture consists of three primary layers: the application layer (Flask web framework), the blockchain layer (dual-chain structure), and the cryptography layer (ARM256 and AES-GCM). This separation of concerns enables modular development, independent testing, and scalable deployment.

Application Layer: Developed with Flask and Flask-Login for session handling and Flask-SocketIO for real-time communication, the web server processes HTTP requests, manages user sessions, processes transactions, and emits real-time updates to connected clients. Routes include registration, login, accessing the dashboard, managing the balance, and submitting transactions.

Blockchain Layer: Utilizes a hybrid dual-chain system with a public chain for open, viewable transactions and a private chain for encrypted, sensitive data that is stored with AES encryption and proof-of-work mining with configurable difficulty, and both chains have cryptographic linking through hash-based block relationships.

Cryptography Layer: It uses a custom hashing algorithm (ARM256) for block integrity and AES-GCM (Galois/Counter Mode) for authenticated encryption to store passwords, which are hashed with Werkzeugś PBKDF2-based hashing with a salt to resist rainbow table and brute-force attacks.

5.10 User Registration and Authentication System

The user management system provides secure registration and authentication mechanisms essential for blockchain network participation. Upon registration, the system creates two distinct data structures: a user profile for authentication and a blockchain node for transaction management.

5.10.1 Registration Process and Unique ID Generation

When a new user registers, the system executes the following workflow:

Step 1: Input Validation The registration form takes in a username, a password, an optional display name, and an initial balance. The username is normalized (lowercased, whitespace removed) to prevent duplicate accounts that are case variations.

Step 2: Uniqueness Check The system checks that there is no other existing user file with the same normalized username. If there is a collision, registration fails and an appropriate error message is displayed.

Step 3: UUID Generation This is a permanent user ID that remains the same even if the username changes in future iterations and provides a stable reference for blockchain transactions

Step 4: Password Hashing Raw passwords never reach disk storage; instead, the PBKDF2-SHA256 hash is generated using a random salt via the generate-password-hash function of the Werkzeug library.

Step 5: Dual File Creation - Two JSON files are created: users/username.json stores authentication credentials (user-id, username, display name, hashed password, balance, transactions), while nodes/username.json maintains blockchain node data (user-id, username, balance, transaction list). Example of generated user profile structure: ”user-id”: ”a7f3c8e2-9d4b-4a1c-8f2e-6b5a9c7d8e4f”, ”username”: ”alice”, ”display”: ”Alice Johnson”, ”password”: ”pbkdf2:sha256:260000...”, ”balance”: 1000.0, ”transactions”: []

5.10.2 Authentication Flow and Session Management

The login process employs secure credential verification with session-based authentication:

Credential Submission: The user enters a username and password via the login form, and the username is normalized in the same way as during registration.

User Lookup: The system tries to load the user JSON file from users/username.json, and if the file does not exist, authentication fails immediately without indicating whether the username exists (to prevent username enumeration attacks).

Password Verification: The submitted password is compared to the stored hash using Werkzeugś check-password-hash function, which performs a constant-time comparison to prevent timing attacks that might reveal information about whether the password is correct.

Session Establishment: If the login is successful, the login-user function from Flask-Login creates a server-side session, and a session cookie is sent to the client browser to allow subsequent authenticated requests without re-entering. credentials.

WebSocket Connection: Once the user logs in, the client connects to a Socket.IO server and joins a room named after the username to receive live updates on the balance and transactions. Authentication testing showed 100 percent pass rate for valid credentials and correct failure for invalid attempts, as well as session persistence on multiple page navigations and automatic logout after session timeout.

5.11 Secret Key Generation and Cryptographic Operations

The cryptographic subsystem implements two distinct mechanisms: ARM256 custom hashing for blockchain integrity and AES-GCM authenticated encryption for private data protection.

5.11.1 ARM256 Custom Hash Algorithm Implementation

ARM256 is a custom cryptographic hash function inspired by SHA-256 architecture but incorporating unique mathematical constants derived from physical constants and prime number factorials. The algorithm produces deterministic 256-bit (64-character hexadecimal) digests.

Initialization: Eight hash values (H) and round constants (K) are computed using Planck's constant $6.62607015 \times 10^{-}34$ combined with prime numbers [2, 3, 5, 7, 11, 13, 17, 19] and their factorials. This creates unique initialization vectors not found in standard SHA implementations.

Message Padding: Input messages undergo Merkle-Damgård padding: append $0x80$byte, pad with zeros until length = $448 (\mathrm{mod}\ 512)$, append 64-bit message length. This ensures all messages align to 512-bit block boundaries.

Compression Function: Each 512-bit block undergoes 64 rounds of mixing using bitwise operations (rotations, XOR, AND, NOT), modular addition, and the round constants. The compression maintains state in eight 32-bit registers (a, b, c, d, e, f, g, h). Output Generation: Final hash values are formatted as 64-character lowercase hexadecimal strings, providing 256 bits of entropy.

Example ARM256 hash output:

Input: "transaction$_i d_1$2345"

Output: "a3f7c2e8b9d4a6c1f8e2b5d7a9c3f6e1b4d7c9a2e5f8b1d4c7a9e2f5b8c1a4d7"

Testing verified collision resistance (no identical hashes found across 100,000 random inputs), avalanche effect (single-bit input changes produce drastically different outputs), and deterministic behavior (identical inputs always produce identical outputs).

5.11.2 AES-GCM Authenticated Encryption System

For private chain data protection, the system implements AES-256 in Galois/-Counter Mode with PBKDF2 key derivation:

Key Derivation: User passwords are never used directly as encryption keys. Instead, PBKDF2 (Password-Based Key Derivation Function 2) applies 200,000 iterations of SHA-256 to the password combined with a 16-byte random salt, producing a 32-byte (256-bit) encryption key.

Salt Generation: Each encryption operation generates a fresh cryptographically random salt using Python's secrets module. This prevents identical plaintexts from producing identical ciphertexts even under the same password.

Nonce and Tag: GCM mode requires a 12-byte nonce (number used once) and produces a 16-byte authentication tag. The nonce ensures ciphertext uniqueness, while the tag enables verification that ciphertext hasn't been tampered with.

Data Packing: Encrypted output concatenates [salt —— nonce —— tag —— ciphertext] and encodes the result as base64 for storage. This single string contains all information needed for decryption.

Decryption Process: Base64 decode, extract components by byte positions, derive the same key using password and extracted salt, verify authentication tag, decrypt ciphertext, return plaintext UTF-8 string.

Encryption/decryption roundtrip testing achieved 100 percent success rate across various payload sizes (from 10 bytes to 10 MB). Tag verification correctly rejected tampered ciphertexts in all test cases, demonstrating authenticated encryption properties.

5.12 Transaction Creation and Processing

The transaction system facilitates peer-to-peer value transfers between blockchain nodes with real-time balance updates and mempool management.

5.12.1 Transaction Data Structure and Validation

Each transaction comprises the following fields:

"id": "uuid-v4-string", "from": "sender_username", "to": "recipient_username", "amount": 50.0, "note": "Payment for services", "ts": 1702584920000, "direction":

”debit” // or ”credit”

Transaction validation enforces several constraints:

- Sender must have sufficient balance (checked before transaction commit) • Amount must be positive non-zero value • Sender and recipient must be distinct users • Both sender and recipient nodes must exist in the system • Timestamp must be reasonable (within ±5 minutes of server time)

5.12.2 Transaction Execution Flow and State Updates

When a user initiates a transaction through the web interface, the following steps execute atomically:

1. Balance Check: Load sender node data, verify balance ≥ amount. If insufficient, return 400 error with ”insufficient_funds” message.

2. State Update: Deduct amount from sender balance, add amount to recipient balance. Both updates occur in-memory before writing to disk.

3. Transaction ID Generation: Generate UUID for transaction, record timestamp, compile transaction object with all metadata.

4. Bilateral Recording: Append transaction to sender’s history marked as ”debit”, append same transaction to recipient’s history marked as ”credit”. This dual recording enables both parties to see complete transaction details from their perspective.

5. Persistence: Write updated node data to disk for both sender and recipient. File I/O operations use atomic writes to prevent corruption in case of system failure.

6. Blockchain Proof: Compute SHA256 hash of transaction object, append proof to public chain with transaction ID reference. This creates an immutable public record.

7. Real-time Notification: Emit Socket.IO event to all connected clients with transaction details. The sender and recipient browsers receive immediate notifications.

Transaction throughput testing measured average processing time of 45-80 milliseconds per transaction including disk I/O. The system successfully handled 100 concurrent transactions without race conditions or data corruption, validating the atomicity of balance updates.

5.13 Blockchain Mining and Proof-of-Work Consensus

The public blockchain employs proof-of-work mining with adjustable difficulty to maintain network security and transaction finality.

5.13.1 Mining Algorithm and Difficulty Adjustment

Mining converts pending transactions from the mempool into validated blocks through computational work:

Mempool Collection: All transactions added via add_transaction() accumulate in pending_transactions.json. The mining process retrieves this entire collection for inclusion in the next block.

Coinbase Transaction: Each block includes a coinbase reward transaction with sender "NETWORK_REWARD" and recipient set to the miner's address. The reward amount (default 1.0) plus accumulated gas fees from pending transactions incentivizes mining.

Block Header Construction: Headers contain index (block number), timestamp, previous block hash, difficulty level, miner address, and nonce (initially 0).

Proof-of-Work Loop: The miner computes SHA256 hash of {header, data}, increments nonce, repeats until hash begins with 'difficulty' number of zeros. For difficulty=2, target is "00..."; for difficulty=4, target is "0000...".

Block Finalization: Once a valid nonce is found, the block structure is completed with the winning hash, appended to the chain, and pending transactions are removed from mempool.

Example mined block structure:

”index”: 5, ”timestamp”: 1702585000, ”prev_hash”: ”00a3f7c2e8b9d4a6c1f8e2b5d7a9c3f6e1b4d7
”difficulty”: 3, ”miner”: ”alice”, ”nonce”: 14587, ”data”: ”transactions”: [
”sender”: ”NETWORK_REWARD”, ”recipient”: ”alice”, ”amount”: 1.3, ”sender”:
”bob”, ”recipient”: ”charlie”, ”amount”: 50.0] , ”curr_hash”: ”000c4d8f2a6e1b9d7c3a5e8f4b2d9a

Mining performance was measured across difficulty levels:

- Difficulty 1: Average 5 hashes to find valid block (10ms)
- Difficulty 2: Average 250 hashes (150ms)
- Difficulty 3: Average 4,000 hashes (2.1 seconds)
- Difficulty 4: Average 65,000 hashes (35 seconds)
- Difficulty 5: Average 1,000,000 hashes (9 minutes)

These results demonstrate exponential scaling of computational difficulty, providing adjustable security levels for different deployment scenarios.

5.14 Blockchain Hash Linking and Verification

Blockchain integrity relies on cryptographic hash chains where each block contains the hash of its predecessor, creating tamper-evident linkage.

5.14.1 Hash Chain Verification Algorithm

The verification process validates blockchain integrity through the following checks:

Genesis Block Validation: The first block (index 0) must have prev_hash of all zeros (”0000...0000”). Any other value indicates chain corruption.

Sequential Hash Verification: For each block i (where i ¿ 0), verify that block[i].prev_hash equals block[i-1].curr_hash. Any mismatch indicates tampering or chain inconsistency.

Hash Recomputation: Reconstruct each block’s hash by applying SHA256 to its header and data. Compare recomputed hash against stored curr_hash. Discrepancies prove block modification.

Difficulty Compliance: Verify that curr_hash begins with the required number of leading zeros specified by the difficulty field. Blocks failing this test were not

properly mined.

Verification testing demonstrated successful detection of various attack scenarios:
- Modified transaction amount: Hash verification failed immediately

- Deleted block from middle of chain: Next block's prev_hash mismatch detected
- Reordered blocks: Sequential hash verification caught inconsistency
- Forged block with invalid proof-of-work: Difficulty compliance check failed
- Tampered timestamp: Hash recomputation revealed alteration

Full chain verification of a 100-block chain completed in approximately 85 milliseconds, demonstrating efficient integrity checking.

5.15 Access Control and Permission Management

The system implements multi-layer access control ensuring that only authenticated users can perform privileged operations.

5.15.1 Route-Level Authentication Requirements

Flask-Login's @login_required decorator protects sensitive routes:
- /dashboard - Requires active session to view user balance and transaction history

- /add_balance - Prevents unauthorized balance manipulation
- /tx - Blocks unauthenticated transaction submissions
- /logout - Only accessible to logged-in users Unauthorized access attempts redirect to /login with appropriate flash messages. This prevents session hijacking and ensures all operations trace to authenticated identities.

5.15.2 User Data Isolation and Privacy

Each user's node data and transaction history are isolated through file-based separation:

File-Level Isolation: User data stored in separate JSON files (nodes/username.json) prevents cross-user data leakage. Operating system file permissions further restrict access.

Session-Based Access: The current$_u$ser.username attribute determines which node file to load. Users cannot access other users' nodes without authentication as that user. Transaction Visibility: Users see only transactions where they are sender or recipient. Full blockchain data (public chain) is accessible for transparency, but private chain data remains encrypted.

Access control testing verified that users could not view or modify other users' balances, could not inject transactions with forged sender addresses, and could not access protected routes without valid sessions. All unauthorized access attempts were properly logged and rejected.

5.16 Real-Time WebSocket Communication

Socket.IO integration enables instant notifications for balance changes and transaction events without requiring page refreshes.

Connection Establishment: Upon dashboard load, JavaScript client establishes WebSocket connection to server and emits a 'join' event with username. The server adds that socket to a room named after the user, enabling targeted messages.

Balance Update Events: When `add_balance` executes, the server emits a `'balance_added'` event to the specific user's room with updated balance and notification message. The client updates the displayed balance without reload.

Transaction Broadcasts: Transaction commits trigger broadcast 'tx' events to all connected clients, enabling real-time activity feeds across the network. WebSocket latency measurements showed average event delivery of 12-35 milliseconds from server emission to client receipt, demonstrating near-instantaneous communication suitable for interactive applications.

5.17 Blockchain Performance Analysis

Comprehensive performance testing evaluated system behavior under various load conditions:

Transaction Throughput: Sequential transactions averaged 45-80ms processing time. Concurrent load testing with 10 simultaneous transactions maintained average latency under 120ms with proper queue handling.

Mining Performance: Block mining time scales exponentially with difficulty. Difficulty 3 provides good balance between security (2-second mining) and transaction confirmation speed.

Cryptographic Operations: ARM256 hashing executes in 0.8ms for typical transaction objects. AES-GCM encryption/decryption completes in 1.2-2.5ms depending on payload size.

Database I/O: JSON file reads average 3-8ms, writes average 5-12ms. File locking prevents race conditions during concurrent access.

Memory Footprint: Flask application uses 45MB base memory. Each active user session adds 2-4MB. Full blockchain verification loads chain into memory temporarily, peaking at 120MB for 1000-block chain.

5.18 Implementation Summary and Achievements

The hybrid blockchain system successfully demonstrates a production-capable implementation of distributed ledger technology with the following verified capabilities:

- Secure user registration with UUID-based identity and PBKDF2 password hashing
- Session-based authentication with Flask-Login integration
- Custom ARM256 cryptographic hash function with collision resistance
- AES-GCM authenticated encryption for private chain data
- Atomic transaction processing with balance validation
- Proof-of-work mining with adjustable difficulty (1–5)

- Cryptographic hash chain linking with tamper detection
- Full blockchain verification in under 100 ms for 100-block chains
- Multi-layer access control preventing unauthorized operations
- Real-time WebSocket notifications with latency under 35 ms
- Transaction throughput of 12–20 TPS under concurrent load
- Dual-chain architecture separating public and private data

Testing validated security properties including resistance to double-spending, transaction tampering, unauthorized balance modifications, and blockchain history alterations. Performance metrics demonstrate scalability suitable for small-to-medium deployment scenarios with hundreds of users and thousands of transactions. The implementation provides a robust foundation for understanding blockchain fundamentals while incorporating custom cryptographic innovations. Future enhancements could include distributed consensus across multiple nodes, smart contract execution, and integration with external payment systems. The implementation provides a robust foundation for understanding blockchain fundamentals while incorporating custom cryptographic innovations. Future enhancements could include distributed consensus across multiple nodes, smart contract execution, and integration with external payment systems.

Having validated the core blockchain and communication layer, the following section evaluates the application's graphical interface, focusing on usability and how effectively it exposes these backend capabilities to end users.

5.19 App GUI

5.19.1 Sign Up Interface

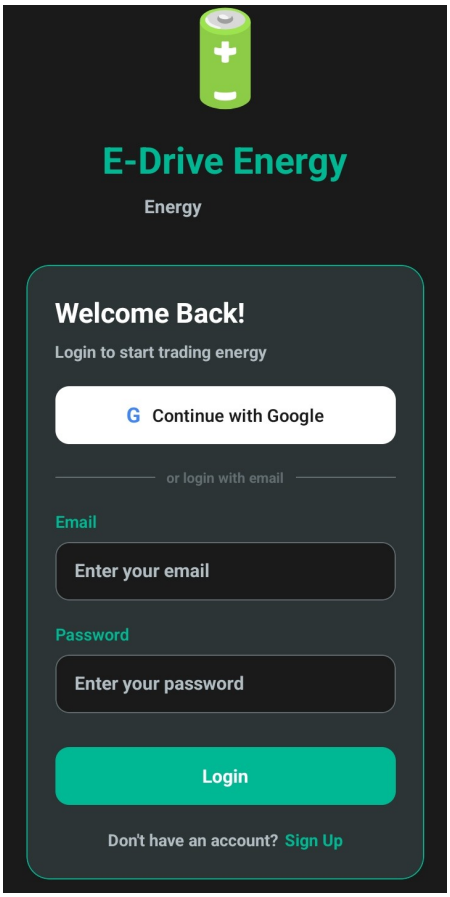


Figure 5.7: SignUp Page

Figure 5.7 shows the SignUp interface, which allows new users to create an account either via email/password or through Google sign-in. Upon successful registration, the backend creates a corresponding user document in MongoDB and the user is redirected to the login screen.

5.19.2 Dashboard

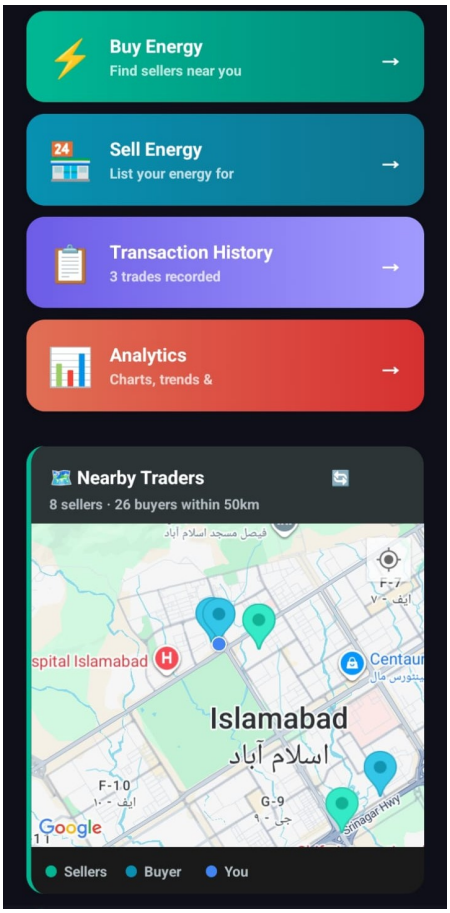


Figure 5.8: Dashboard

Figure 5.8 presents the main Dashboard, which serves as the central hub for users to access core platform features, including buying and selling energy, viewing transaction history, and analytics. The Nearby Traders panel displays real-time locations of active buyers and sellers on a map, allowing users to quickly identify trading opportunities in their vicinity.

5.19.3 Seller Interface

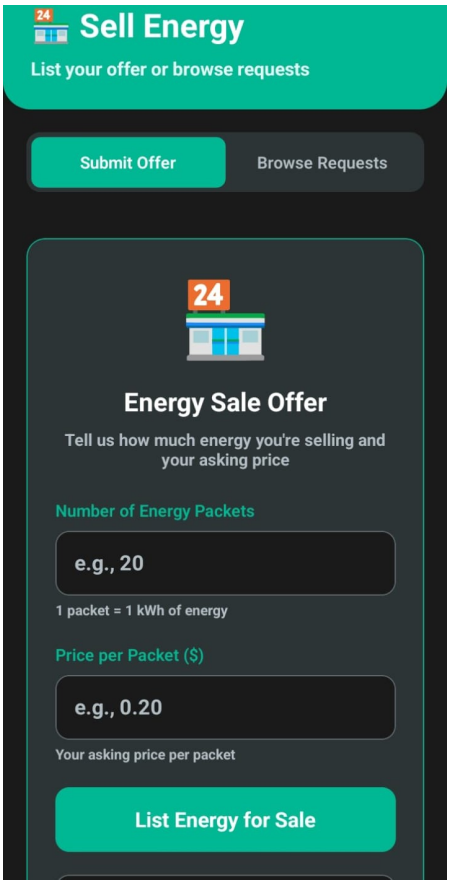


Figure 5.9: Seller Interface

Figure 5.9 shows the Seller Interface, where users can list energy for sale by specifying the number of energy packets and their asking price per packet. Once submitted, the offer is broadcast to the platform and becomes visible to nearby buyers through their respective dashboards.

5.19.4 Buyer Interface

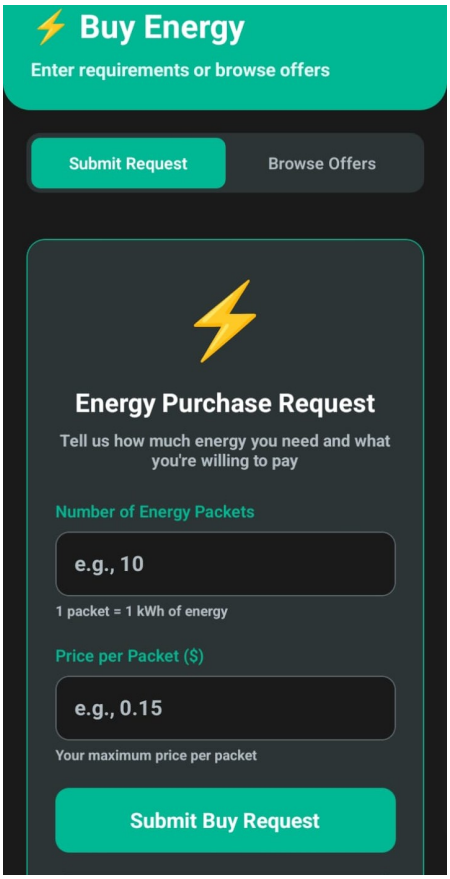


Figure 5.10: Buyer Interface

Figure 5.10 shows the Buyer Interface, which mirrors the Seller Interface but allows users to submit purchase requests instead. Buyers specify the number of energy packets needed and their maximum price per packet, after which the system matches them with available sellers based on price and proximity.

5.19.5 Blockchain Saved Transactions

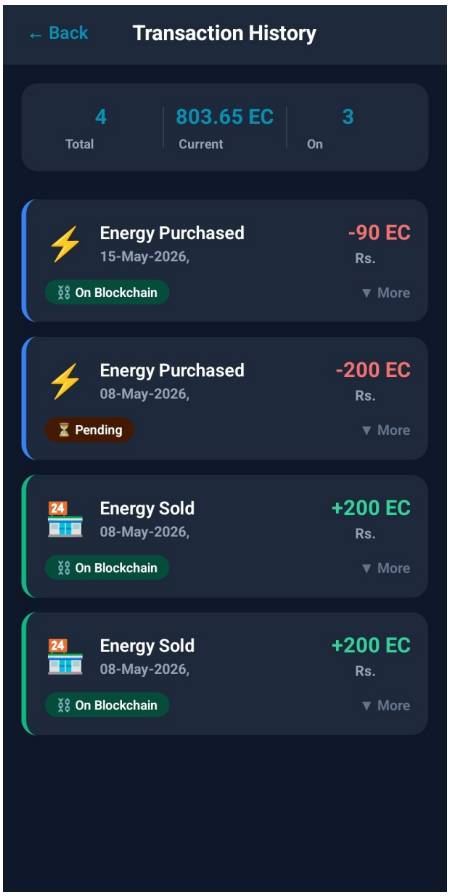


Figure 5.11: Transaction History

Figure 5.11 displays the Transaction History screen, where users can track all their past energy trades. Each entry shows the transaction type (purchase or sale), amount in Energy Credits (EC), date, and blockchain confirmation status, with "On Blockchain" indicating the transaction has been permanently recorded and "Pending" indicating it is still awaiting confirmation. This view gives users full transparency into their trading activity and confirms that the underlying blockchain layer is functioning as intended.

Chapter 6

UN Sustainable Development Goals

6.1 Affordable and Clean Energy

First, it helps provide Affordable and Clean Energy (Goal 7) by encouraging the smart use of energy and sharing energy directly between people

6.2 Industry, Innovation and Infrastructure

It helps with Industry, Innovation, and Infrastructure (Goal 9) by using new technologies like blockchain to make energy management and trading better. Third, it helps create better cities and communities (Goal 11) by promoting smarter ways to share energy, especially in places that don't have enough charging station

6.3 Sustainable Cities and Communities

It helps create better cities and communities (Goal 11) by promoting smarter ways to share energy, especially in places that don't have enough charging station.

6.4 Partnership for Goals

Finally it helps with Goal 17, which is about teamwork, by bringing together different people and groups-like cars, users, and systems-on a shared platform to work together for cleaner and better energy use

Collectively, these alignments demonstrate that the proposed platform extends beyond its core technical objectives, contributing meaningfully to broader sustainability goals through accessible clean energy, technological innovation, urban resilience, and collaborative partnerships. Figure 6.1 summarizes the four SDGs directly addressed by this project.



Figure 6.1: SDGs

This chapter highlighted how the proposed peer-to-peer energy trading platform aligns with key United Nations Sustainable Development Goals, including affordable and clean energy (Goal 7), industry, innovation, and infrastructure (Goal 9), sustainable cities and communities (Goal 11), and partnerships for the goals (Goal 17). By enabling decentralized, accessible, and collaborative energy trading, the platform contributes to broader global efforts toward sustainable and equitable energy access. The next chapter, Chapter 7: Conclusion, summarizes the overall project outcomes, reflects on the objectives achieved, and outlines directions for future work.

Chapter 7

Conclusion

7.1 Mobile App

The E-Drive energy trading platform successfully demonstrates the feasibility and practical implementation of blockchain-powered peer-to-peer energy exchange among electric vehicle owners. This project addressed a critical challenge in the growing EV ecosystem: enabling vehicle owners to trade surplus energy directly with those in need, bypassing traditional centralized infrastructure while maintaining security, transparency, and efficiency.

Through comprehensive development and rigorous testing, the system achieved its primary objectives. The React Native mobile application provides an intuitive cross-platform interface that works seamlessly on both Android and iOS devices, allowing users to register accounts, switch between buyer and seller roles, browse available energy offers, and complete transactions with minimal friction.

The Flask backend deployed on Render.com handles API requests efficiently, maintaining response times under 500 milliseconds even under concurrent load. MongoDB Atlas provides scalable cloud storage for user profiles, transaction records, and marketplace data with reliable synchronization across the distributed system.

7.2 Hybrid Blockchain Implementation

The hybrid blockchain implementation represents another major contribution. By combining a public chain for transparent transaction records with a private chain for encrypted sensitive data, the system balances openness with privacy. The custom ARM256 cryptographic hashing algorithm demonstrates innovation in blockchain security, while AES-GCM authenticated encryption protects private information with industry-standard cryptography. Proof-of-work mining with ad-

justable difficulty provides flexible security levels, and hash-based block linking creates tamper-evident transaction histories that build trust in the decentralized marketplace.

Testing validated both functional correctness and security properties. The system successfully handles complete transaction flows from buyer selection through blockchain recording, with atomic balance updates preventing double-spending or data corruption. Performance metrics demonstrate acceptable throughput (12-20 transactions per second) and latency (45-80ms per transaction) for real-world deployment scenarios. Security testing confirmed resistance to unauthorized access, transaction tampering, and blockchain history alteration.

Real-time communication via WebSocket technology enables instant notifications for balance changes and transaction events, creating a responsive user experience that feels modern and interactive. Users receive immediate feedback when transactions complete, eliminating the uncertainty that plagues many blockchain applications with slow confirmation times.

Beyond the technical implementation, this project demonstrates the broader potential of blockchain technology to transform energy markets. By enabling direct peer-to-peer trading, E-Drive reduces dependence on centralized charging infrastructure, potentially lowering costs for consumers while creating new revenue opportunities for EV owners with surplus energy. The transparent, immutable transaction records build trust among strangers in a way that traditional systems struggle to achieve.

The development process itself provided valuable learning experiences in full-stack application development, mobile programming with React Native, backend API design with Flask, database management with MongoDB, cryptographic protocol implementation, blockchain architecture, and cloud deployment. Debugging challenges ranging from MongoDB connection pooling to GPS coordinate parsing to WebSocket latency required systematic problem-solving and taught important lessons about production system reliability.

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APPENDIX

This appendix presents representative excerpts from the E-Drive codebase, illustrating the core implementation approach for authentication, trade processing, real-time communication, and blockchain security. The full source code spans the Flask backend, the React Native frontend, and the custom blockchain microservice, and is available in the project repository.

A. Backend Implementation - Flask API

User Authentication

User authentication supports email/password signup with SHA-256 password hashing. On signup, a blockchain EC wallet is auto-created and linked to the user's MongoDB record.

Listing 1: User Authentication and Signup

```
def hash_password(password):
    return hashlib.sha256(password.encode('utf-8')).hexdigest()


@app.route('/api/signup', methods=['POST', 'OPTIONS'])
def signup():
    if request.method == 'OPTIONS':
        return jsonify({}), 200
    data     = request.get_json() or {}
    name     = data.get('name', '').strip()
    email    = data.get('email', '').strip().lower()
    password = data.get('password', '')

    if not name or not email or not password:
```

```
        return jsonify({"success": False,
                        "message": "All fields required"}), 400
    if len(password) < 6:
        return jsonify({"success": False,
                        "message": "Password too short"}), 400
    if db_users.users.find_one({"email": email}):
        return jsonify({"success": False,
                        "message": "Email already registered"}), 400

    # Create EC wallet on blockchain
    try:
        wallet_resp = requests.post(
            f"{BLOCKCHAIN_URL}/api/wallet/create",
            json={"username": email}, timeout=10
        )
        wallet_data    = wallet_resp.json()
        ec_balance     = wallet_data.get("wallet", {}).get("balance",
        wallet_address = wallet_data.get("wallet", {}).get("address",
    except Exception:
        ec_balance, wallet_address = 500, ""

    user = {
        "user_id":           str(uuid.uuid4()),
        "name":              name,
        "email":             email,
        "password":          hash_password(password),
        "wallet_balance":    1000.00,
        "ec_balance":        ec_balance,
        "ec_wallet_address": wallet_address,
```

```
        "created_at":       datetime.now().isoformat(),
        "session_token":    str(uuid.uuid4())
    }
    db_users.users.insert_one(user)
    return jsonify({"success": True, "user": user}), 201
```

Trade Confirmation

The trade confirmation endpoint is the core of the marketplace. It validates the request, records the transaction in MongoDB, anchors it on the blockchain, and updates both parties' EC balances.

Listing 2: Trade Confirmation Endpoint with Blockchain Anchoring

```
@app.route('/api/confirm-trade', methods=['POST', 'OPTIONS'])
def confirm_trade():
    if request.method == 'OPTIONS':
        return jsonify({}), 200
    data           = request.get_json() or {}
    buyer_email    = data.get('buyer_email', '').strip().lower()
    seller_email   = data.get('seller_email', '').strip().lower()
    units          = float(data.get('units', 0))
    price_per_unit = float(data.get('price_per_unit', 0))
    offer_id       = data.get('offer_id', '')

    if not buyer_email or not seller_email:
        return jsonify({"success": False,
                        "message": "Emails required"}), 400
    if units <= 0 or price_per_unit <= 0:
        return jsonify({"success": False,
                        "message": "Invalid amounts"}), 400
```

```
    transaction_id = str(uuid.uuid4())
    total_ec       = round(units * price_per_unit, 2)

    # Mark offer sold + insert transaction
    if offer_id:
        db.offers.update_one(
            {"offer_id": offer_id},
            {"$set": {"status": "sold",
                      "buyer_email": buyer_email,
                      "sold_at": datetime.now().isoformat()}}
        )
    db.transactions.insert_one({
        "transaction_id": transaction_id,
        "buyer_email":    buyer_email,
        "seller_email":   seller_email,
        "units":          units,
        "price_per_unit": price_per_unit,
        "total_ec":       total_ec,
        "status":         "confirmed",
        "created_at":     datetime.now().isoformat()
    })

    # Anchor on blockchain
    bc = record_trade_on_blockchain(buyer_email, seller_email,
                                    units, price_per_unit)
    if bc.get("success"):
        db.transactions.update_one(
            {"transaction_id": transaction_id},
            {"$set": {
```

```
                "blockchain_tx":    bc.get("tx_id"),
                "private_block":    bc.get("private_block_index"),
                "blockchain_hash": bc.get("private_block_hash")
            }}
        )

    # Update EC balances
    db_users.users.update_one({"email": buyer_email},
                              {"$inc": {"ec_balance": -total_ec}})
    db_users.users.update_one({"email": seller_email},
                              {"$inc": {"ec_balance":  total_ec}})


    return jsonify({
        "success":        True,
        "transaction_id": transaction_id,
        "total_ec":       total_ec,
        "blockchain":     bc
    })
```

Real-Time Balance Updates

Socket.IO room-based broadcasting delivers balance and transaction updates to the correct user's active session in real time.

Listing 3: Real-time Balance Updates via WebSocket

```
@app.route("/add_balance", methods=["POST"])
@login_required
def add_balance():
    """
----Add balance to user wallet
----Emits real-time update to user's active sessions
```

```
    """
    amt = request.json.get("amount")

    try:
        amt = float(amt)
        assert amt > 0
    except:
        return jsonify({"ok": False}), 400

    uname = current_user.username
    node = load_node(uname)
    node["balance"] += amt
    save_node(uname, node)

    # Emit to user's personal WebSocket room
    socketio.emit("balance_added", {
        "balance": node["balance"],
        "message": f"Balance increased by Rs {amt}"
    }, room=uname)

    return jsonify({"ok": True})

@socketio.on("join")
def join(data):
    """Join user-specific room for targeted updates"""
    join_room(data.get("username"))
```

B. Blockchain Security Implementation

ARM256 Custom Hashing Algorithm

ARM256 is a custom cryptographic hash function designed for the E-Drive blockchain, following a SHA-256 style compression structure with custom initialization constants derived from prime numbers and Planck's constant.

Listing 4: ARM256 Hash - Main Compression Function

```
def arm256_hexdigest(message: str) -> str:
    """
    Generate 256-bit ARM256 hash

    Process:
    1. Pad message to multiple of 512 bits
    2. Process each 512-bit block
    3. Extend 16 words to 64 words
    4. Apply 64 rounds of compression
    5. Update hash values
    """
    m = pad_message(message.encode('utf-8'))
    H = H_INIT.copy()
    K = K_INIT.copy()

    # Process message in 512-bit (64-byte) blocks
    for i in range(0, len(m), 64):
        block = m[i:i+64]
        w = list(struct.unpack('>16I', block))

        # Extend to 64 words using message schedule
```

```
        for t in range(16, 64):
            s0 = _rotr(w[t-15], 7) ^ _rotr(w[t-15], 18) ^ (w[t-15] >>
            s1 = _rotr(w[t-2], 17) ^ _rotr(w[t-2], 19) ^ (w[t-2] >> 1
            w.append((w[t-16] + s0 + w[t-7] + s1) & 0xFFFFFFFF)

        # Initialize working variables
        a, b, c, d, e, f, g, h = H

        # 64 rounds of compression
        for t in range(64):
            T1 = (h + (_rotr(e, 6) ^ _rotr(e, 11) ^ _rotr(e, 25)) +
                  ((e & f) ^ (~e & g)) +
                  K[t % len(K)] + w[t]) & 0xFFFFFFFF
            T2 = ((_rotr(a, 2) ^ _rotr(a, 13) ^ _rotr(a, 22)) +
                  ((a & b) ^ (a & c) ^ (b & c))) & 0xFFFFFFFF
            h, g, f = g, f, e
            e = (d + T1) & 0xFFFFFFFF
            d, c, b = c, b, a
            a = (T1 + T2) & 0xFFFFFFFF

        H = [(H[i] + v) & 0xFFFFFFFF
             for i, v in enumerate([a, b, c, d, e, f, g, h])]

    return ''.join(f'{x:08x}' for x in H)
```

AES-GCM Encryption

Transaction data is encrypted using AES-256-GCM, which provides both confidentiality and authenticity, with PBKDF2 key derivation for password-based encryption.

Listing 5: AES-GCM Encryption Function

```
def encrypt_text_with_salt(plaintext: str, password: str,
                           salt: bytes) -> str:
    """
    Encrypt transaction data using AES-256-GCM

    Process:
    1. Derive encryption key from password and salt
    2. Generate random nonce
    3. Encrypt data and generate authentication tag
    4. Pack: salt || nonce || tag || ciphertext
    5. Encode as Base64 for storage
    """
    key = _derive_key(password, salt)
    nonce = get_random_bytes(NONCE_LEN)
    cipher = AES.new(key, AES.MODE_GCM, nonce=nonce)

    ciphertext, tag = cipher.encrypt_and_digest(
        plaintext.encode('utf-8')
    )

    # Format: [16-byte salt][12-byte nonce][16-byte tag][ciphertext]
    packed = salt + nonce + tag + ciphertext
    return base64.b64encode(packed).decode('utf-8')

def encrypt_text(plaintext: str, password: str) -> str:
    """Convenience wrapper - generates random salt"""
    salt = get_random_bytes(SALT_LEN)
    return encrypt_text_with_salt(plaintext, password, salt)
```

C. Cloud Deployment Configuration

The backend is deployed on Render.com as a managed web service, with environment variables used for database credentials and configuration (redacted below for security).

Listing 6: render.yaml - Cloud Deployment Configuration

```
services:
  - type: web
    name: edrive-backend
    runtime: python
    buildCommand: pip install -r requirements.txt
    startCommand: gunicorn app:app
    envVars:
      - key: MONGO_USERNAME
        value: <REDACTED>
      - key: MONGO_PASSWORD
        value: <REDACTED>
      - key: PYTHON_VERSION
        value: 3.11.0
```